Assessment Summary

To pass this subject you must

Item	Weight	Due Date	SLOs Assessed	Mandatory	Threshold
Presentation 1 & 2	30%	Tutorial time in **Week 5 or 6** and **Week 10 or 11**	1	**YES**	No
Report	40%	**Friday, 16 October, 11:59 PM**	1, 2	**YES**	No
Reflection	30%	**Friday, 23 October, 11:59 PM**	3	**YES**	No

Length: Minimum 12,000 words for 6 member groups (excluding appendixes and any references) Estimate 2000 words per member.

Submission: Turnitin as a Word document.

In your report, each student must demonstrate the process used to develop their project, not only the final outcomes.
Students are expected to provide evidence of their use of BIM authoring tools, design, cost estimation tools, time and scheduling tools (4D), and other relevant digital engineering platforms (e.g., Autodesk Forma, Revit, Navisworks, Civil 3D, InfraWorks, or similar).
Groups should also demonstrate how they collaborated throughout the project, using digital collaboration platforms such as Autodesk Forma Boards.

The report should be written in third-person language, maintaining an objective and professional tone throughout.

The submission is to be in the format of a report with appropriate headings and sub-headings etc.
Identify sources of all information used in your report with consistent Harvard WSU referencing style.
Use relevant images and acknowledge the source, label with image captions and cross reference the image in-text.

Note: Each student must write his ,her or their name and student number next to the section they have contributed to in this report. This will help clearly identify individual contributions.

Links to presentations 1 and 2 for reference
complex building project p2

MAKE SURE ALL FIGURES CORRESPONDED TO EACH SECTION BEFORE SUBMISSION
complex building project p1

Executive Summary

1 Description

turnover of under $ 500 million per annum
Our team has selected Westmead Station as a potential air-space development opportunity to provide Build-to-Rent accommodation for essential workers within one of NSW's major health and innovation precincts. The proposal combines residential accommodation with health-and-wellbeing, office, dining and retail uses while maintaining heavy-rail operations throughout construction.

2 Your team and project

ADDPI Developments - Australian Design, Development, Projects & Infrastructure - NSW Tier 2 Construction Contractor

The team consists of 5; David Azmy, Alyta Huynh, Philip Semenov, Dom Ishak and Ivan Le.

Criteria found from your two presentations:
[Note: For references, students may use both academic and industry-standard sources. They are also encouraged to support their arguments with screenshots from their designs, plans, or any documents they have used. If they believe additional information is important but not essential to the main body, they may include it in the appendix.]

3 Compliance Officer (Dom Ishak)

3.1 Compliance Officer's Role and Responsibilities

The job of the compliance officer is to ensure that the proposed Westmead Station mixed use development meets the relevant planning, building and regulatory requirements throughout the project. The development will consist of constructing residential and commercial facilities near an operational railway station, compliance must be considered from the early design stages through to construction and completion.

The role will have to identify statutory approval requirements, coordinating with relevant authorities and ensuring that the proposed development addresses the National Construction Code (NCC), State Significant Development planning requirements and Transport for NSW requirements. Further attention has to be placed on infrastructure services, fire safety, accessibility and potential risks associated with construction near an operating railway. The compliance officer will be responsible for establishing a compliance management process that identifies regulatory obligations, tracks approval requirements and supports coordination between the design and construction teams. This will help minimise approval delays, design changes and potential compliance issues during construction.

3.2 Statutory Approvals and State Significant Development

The proposed Westmead Station mixed use development requires careful consideration of statutory planning requirements due to its location above an operational railway station and its combination of residential and commercial facilities. Compliance Officers will have to to identify the appropriate development approval pathway and ensure that the project addresses relevant NSW planning legislation before construction begins.

The State Significant Development (SSD) pathway has been identified by the project team as the proposed planning approach. SSD applications are assessed under the *Environmental Planning and Assessment Act 1979*, with the NSW Department of Planning, Housing and Infrastructure coordinating the assessment process. However the project's eligibility for SSD classification must first be confirmed against the relevant planning instruments and development thresholds that being the NSW Department of Planning Housing and Infrastructure.

The SSD approval process will begin when the secretary's environmental assessment requirements are approved and granted. An environmental impact statement is then prepared to assess the potential impacts of the development, including construction noise, traffic, environmental effects, community impacts and infrastructure requirements. For the Westmead Station proposal, an extra compliance consideration is the relationship between the proposed building and the existing railway infrastructure. The State Environmental Planning Policy (Transport and Infrastructure) 2021 contains provisions relating to development near railway tracks and stations, including requirements intended to protect rail operations and public safety. Early communication with Transport for NSW and the relevant rail authorities and council is

therefore important to identify structural, operational and access restrictions before finalising the design (NSW Department of Planning, Housing and Infrastructure 2026).

The Compliance Officer would coordinate these planning requirements through an approval register, recording the relevant authority, required documentation, submission deadlines and outstanding conditions. This process would allow the project team to identify potential approval delays early and coordinate compliance requirements with the design and construction programme.

3.3 National Construction Code (NCC/BCA) Fire Safety and Accessibility Requirements

The proposed Westmead Station development must comply with the National Construction Code (NCC), which establishes minimum requirements for building safety, health, accessibility and performance. As the project includes residential apartments, commercial offices and retail spaces, different building classifications must be considered. These are expected to include Class 2 for residential apartments, Class 5 for offices and Class 6 for retail and hospitality areas (Australian Building Codes Board 2022).

The team recognises that fire safety is particularly important due to the building's height, mixed-use design and connection to an operational railway station. Fire safety planning will involve reviewing fire-resistant construction, compartmentation between different building uses, emergency exits and fire detection and sprinkler systems. Early coordination with fire safety consultants and relevant authorities will help ensure that these requirements are incorporated into the design rather than need costly modifications and changes during construction.

Accessibility is another key consideration, particularly because the development will accommodate residents, workers, customers and railway passengers. The design must address accessible entrances, lifts, circulation and evacuation areas as well as sanitary facilities that comply with NCC Volume One and relevant accessibility standards, including AS 1428.1 (Australian Building Codes Board 2022).

The team aims to identify potential compliance issues during design development and resolve them before construction. This will involve regular design reviews, coordination with specialist consultants and maintaining documentation demonstrating compliance with applicable building requirements. By addressing these matters early, the team aims to minimise approval delays, reduce construction variations and provide the client with greater confidence that the development can be delivered safely and in accordance with NCC and BCA.

3.4 Infrastructure and Services Coordination

Infrastructure and services coordination is a major consideration for the proposed Westmead Station development due to its location within an existing transport precinct. The development will require reliable sewer, water, electricity and stormwater services to support the residential apartments, commercial tenancies and retail facilities. The team aims to ensure that these services are properly assessed and coordinated during the design stage to minimise disruption to existing railway operations and surrounding infrastructure.

Team members will have to identify existing underground services and determine whether the available infrastructure can accommodate for the proposed development. Sydney Water requires developers to obtain approval when construction or excavation may affect existing water and wastewater assets. This may involve reviewing service locations, connection capacity and requirements for protecting or relocating existing infrastructure (Sydney Water 2026). The team will engage Orion as the Water Servicing Coordinator to manage the application and obtain the Section 73 Compliance Certificate from Sydney Water. Orion will assist with reviewing the proposed water and sewer connections, identifying any infrastructure upgrades or protection works required and coordinating the necessary documentation with Sydney Water.

Electrical services must also be considered early, particularly due to the building's residential and commercial demands. The team will coordinate with Endeavour Energy, the electricity distribution network provider servicing the Westmead area, to assess the proposed electrical connections and determine whether additional infrastructure or network upgrades are required. Stormwater management will also require careful planning to ensure that runoff from the development is appropriately collected, controlled and discharged without negatively affecting the railway or surrounding properties.

To manage these requirements the team proposes coordinating hydraulic, electrical and civil engineering designs alongside the architectural and structural plans. Potential conflicts between existing and proposed services will be identified through design reviews and BIM coordination thus allowing issues to be addressed before construction has begun. This will also benefit clients by providing greater certainty regarding infrastructure requirements and associated costs. By resolving service connections, approvals and potential clashes during the design stage, the team aims to reduce unexpected variations and minimise delays.

3.5 BIM Coordination and TFNSW Digital Engineering Framework

Building Information Modelling (BIM) will play an important role in coordinating the proposed Westmead Station development, particularly due to the complexity of constructing a mixed-use building above an operational railway. The team aims to use digital modelling to identify potential design conflicts early, improve communication between consultants and ensure that the proposed building services do not interfere with existing railway infrastructure.

During the design development stage, the team used Autodesk Forma for concept planning and ArchiCAD to develop the building's architectural model. As the project progresses, coordination between architectural, structural and building services models will be required. Software such as Revit and Navisworks can be used to review these models and identify clashes involving structural elements, electrical services, hydraulic pipes and mechanical ventilation systems. Identifying these issues before construction can reduce unnecessary redesign, delays and additional costs. The team will also consider the Transport for NSW digital engineering framework, which promotes consistent management and sharing of digital project information throughout the design and construction process. For the Westmead development, BIM coordination will be particularly important when reviewing structural connections, underground services and areas close to existing railway assets.

The team proposes conducting regular model reviews, recording identified clashes and assigning responsibility to the relevant consultants. Updated models will then be reviewed to confirm that identified issues have been addressed.

3.6 GC21, Safety in Design and Railway Interface Risks

Managing construction risks is an important responsibility for the team, particularly due to the proposed development's location above an operational railway station. The project presents additional safety and coordination challenges, including restricted site access, structural works near railway infrastructure and the need to maintain safe passenger movement throughout construction. The team will also consider the NSW Government's GC21 General Conditions of Contract as a potential framework for managing contractual responsibilities and construction risks. GC21 promotes cooperation between project participants, clear communication and effective management of contractual issues (NSW Government 2022). If put in place this approach would assist the team in managing design changes, contractor responsibilities and potential disputes throughout the project.

Safety within the design will also form an important part of the compliance strategy. Under the Work Health and Safety Act 2011 (NSW) relevant duty holders must identify and manage foreseeable risks associated with the design and construction of structures. The team proposes conducting regular safety reviews with structural engineers, designers and construction representatives to identify hazards before work begins. One of the project's main compliance risks is the interface between the proposed building and existing railway operations. Construction activities such as piling, crane operations and structural works could potentially affect railway assets or passenger safety. To manage these risks, the team will coordinate with

Transport for NSW and the relevant rail authorities to establish appropriate protection measures, work restrictions and approval requirements before construction. For the client, identifying and managing these risks early will provide greater certainty regarding construction safety, programme requirements and potential additional costs.

4 Design Manager (Ivan Le)

4.1 Project Initiation and Feasibility

This project involves constructing a 15-storey mixed-use building directly above Westmead Station, with commercial uses across the lower four levels and residential apartments on the levels above. At initiation, the Design Manager reviewed the site context, including its location within the Westmead health and education precinct, proximity to the hospitals and university, and the constraints of building over a live rail corridor. Feasibility work covered planning controls (height, floor space ratio, setbacks and overshadowing), rail safety and operational requirements, structural transfer demands, vertical separation between station and occupied floors, and access and egress. Early risks were identified and recorded, including construction staging over an operating station, acoustic and vibration impacts on residential levels, fire engineering, and stakeholder approvals from the transport authority. The outcome was a defined design brief, a preliminary space allocation between commercial and residential components, and confirmation that the proposal was viable to proceed to concept design.

4.2 Concept Design (Autodesk Forma)

Concept design began by translating the feasibility findings into testable massing options in Autodesk Forma. The site boundary, the rail corridor and the surrounding buildings were set up as the context model, so that every option was assessed against real conditions rather than in isolation. The 62 m x 41 m floor plate, the 15-storey height and the split between four commercial levels and eleven residential levels were set as the fixed parameters, and the variables tested were the building's position on the site, the podium and tower relationship, and the arrangement of entries.

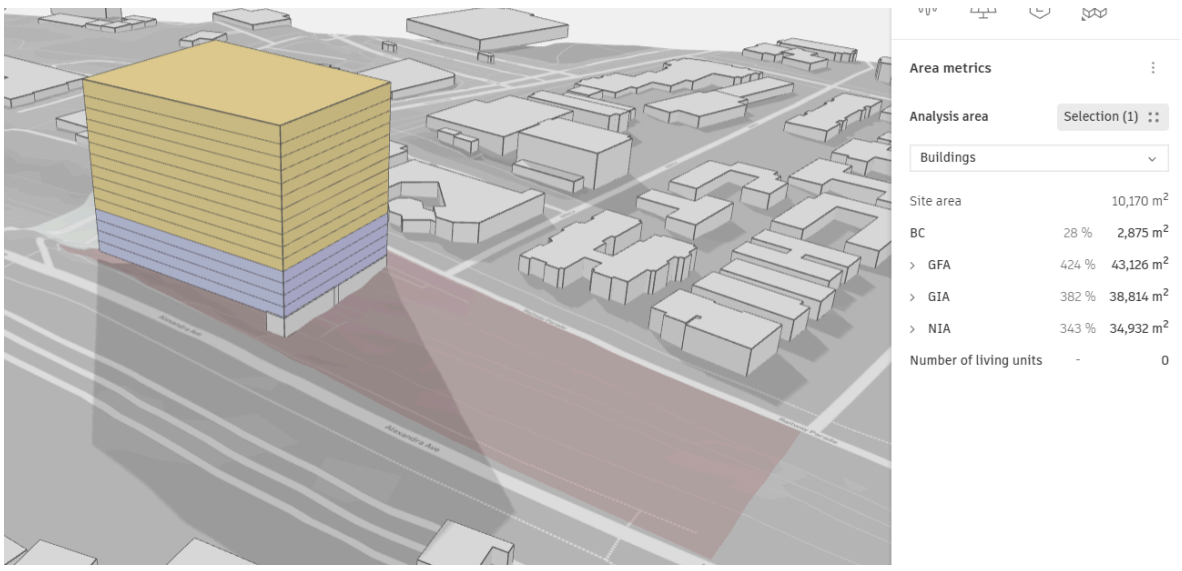


Figure 4.2.1 *Basic site concept with measurements via AutoDesk Forma*

Forma's analysis tools were used to compare the options on environmental performance. Sun-hours and daylight analysis tested how much light reached the residential levels and the neighbouring buildings. Because the two apartments on each residential floor face east and west, the solar analysis was also used to confirm that both balconies receive adequate sun and that neither apartment is disadvantaged by orientation.

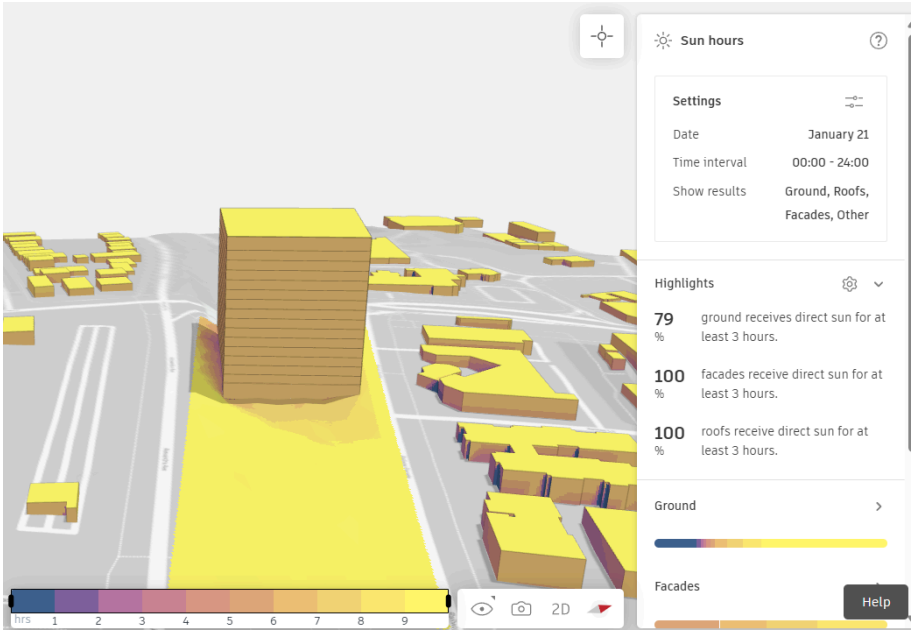


Figure 4.2.2: *Forma analysis results Jan 21 (Summer east-side)*

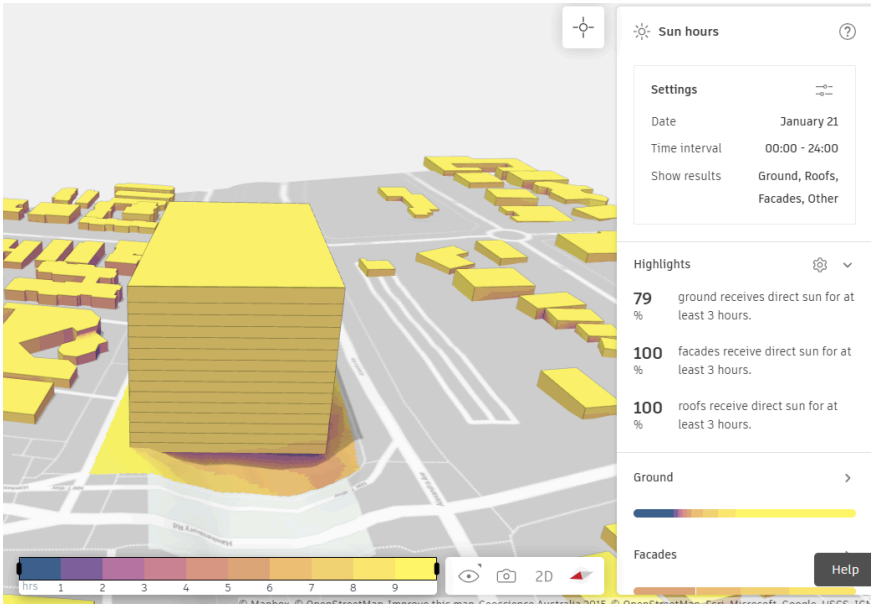


Figure 4.2.3: *Forma analysis results Jan 21 (Summer west-side)*

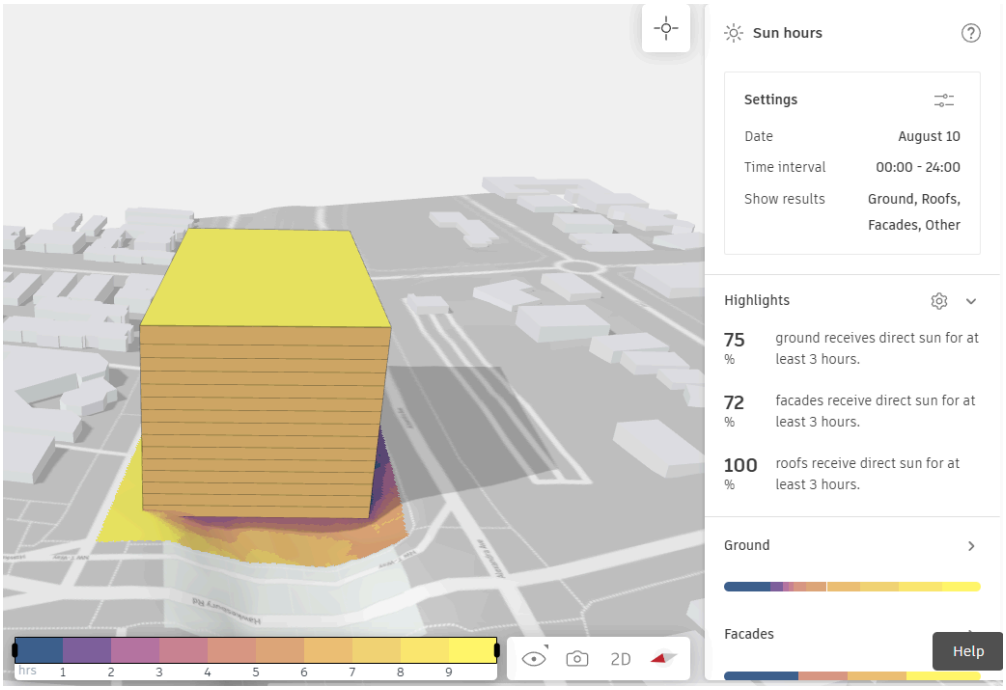


Figure 4.2.4: *Forma analysis results Aug 10 (Winter west-side)*

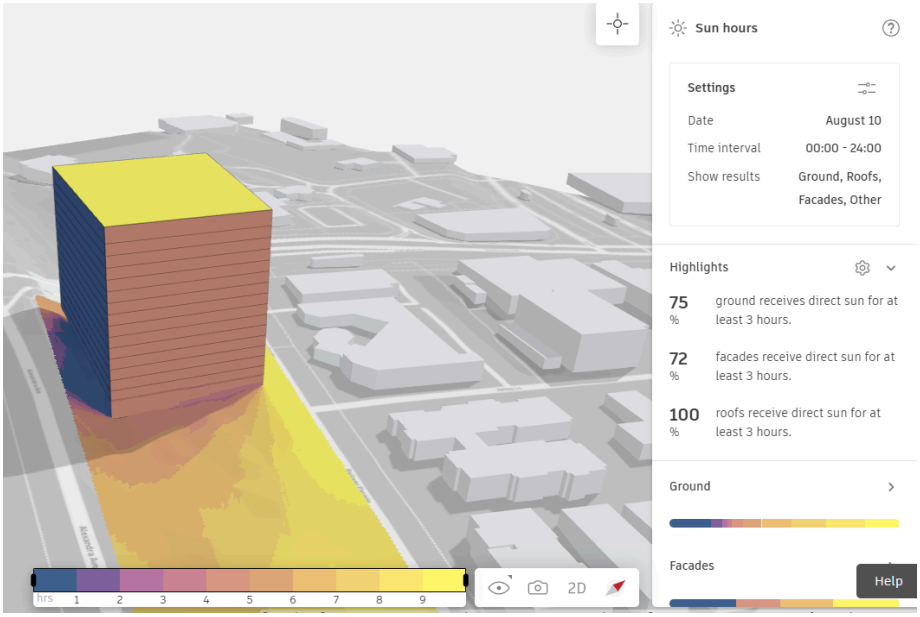


Figure 4.2.5: *Forma analysis results Aug 10 (Winter east-side)*

The options were assessed against the planning controls, the structural and rail constraints from feasibility, and the project objectives. The preferred scheme places the four commercial levels as a podium with active frontages and direct, legible connections to the station concourse. It places the residential levels above, with recessed balconies (shown later in Archicad models) that provide private open space while shading the facade and reducing rail noise intrusion. Both the design and the compliance requirements were reviewed with the Compliance Officer, and the Estimator provided early feedback on the cost implications of the transfer structures and the repeated tower floor plate.

4.3 Schematic Design (Graphisoft ArchiCAD)

Schematic design developed the approved concept into a coordinated BIM model in Graphisoft ArchiCAD. The massing was used to conceptualise the project's storey levels, grids and building envelope, and the walls, slabs, cores, stairs and openings were then modelled on each level. Working in a single model meant that plans, sections and elevations were all generated from the same source, so changes made in one view updated everywhere.

[Figure 4.3.1: ArchiCAD 3D model view of the building]

The commercial podium was planned as four levels with flexible floor plates. The ground floor contains open-fronted cafes and stores, central lift access and toilets, and a further level is dedicated to retail. Their planning was driven by pedestrian flow from the station concourse and by the servicing, loading and fire egress requirements.

[Figure 4.3.2: ArchiCAD ground floor plan]

The residential levels were designed as a single typical floor and repeated across levels 4 to 15. Each floor holds two apartments, each with its own entrance, kitchen, bathroom or ensuite, bedrooms and balcony. The balconies are recessed into the building footprint, and the apartments face east and west. The floor retains a single stair and the west-side lifts, with the unnecessary east-side lifts, refuse, riser and second stair removed to give the apartments more usable area. The apartment entrances and the surrounding corridor were adjusted so that each apartment has a clear, private entry.

[Figure 4.3.3: ArchiCAD typical residential floor plan]

Sections and elevations were produced to check floor-to-floor heights, the podium-to-tower relationship and the façade treatment. The model was also used to check the structural grid against the cores and the station below, to coordinate the location of fire stairs and risers, and to confirm accessibility compliance. Where the structural, services or fire consultants raised issues, the model was updated and the changes were recorded so that the team always worked from a current version.

[Figure 4.3.4: ArchiCAD section or elevation]

4.4 Finalised Building Design

The finalised design is a 15-storey mixed-use building above Westmead Station, with four commercial levels at the base and eleven residential levels above containing 22 apartments in total. The design meets the brief, the planning and building code requirements identified at feasibility, and the operational needs of the station, with clearly separated circulation for rail users, commercial visitors and residents.

[Figure 4.4.1: Final rendered or 3D view of the building]

The final documentation package consists of coordinated floor plans, sections and elevations generated from the ArchiCAD model, along with the model itself, which was reviewed with the Compliance Officer, Estimator, Project Manager, Site Manager and Community Liaison Officer. The review confirmed that the design is compliant, buildable, within the cost expectations and acceptable to the community. Items that remain open, such as final structural transfer details and acoustic treatment, are recorded for the next stage.

[Figure 4.4.2: Final floor plan set or drawing sheet]

The Design Manager handed the finalised design to the Estimator and Project Manager to inform tendering, programming and delivery, and to the Site Manager to plan the staging of construction over the operating station.

5 Estimator (Alyta Huynh - 22062420)

5.1 Estimator Role and Cost Planning Objectives

The Estimator was responsible for developing and progressively refining the financial strategy for the proposed Westmead Station Air-Space Development. The role extended beyond preparation of a single construction estimate and incorporated preliminary cost planning, quantity take-off, benchmarking, escalation, risk allowances, value management and evaluation of the financial implications of design, staging and compliance decisions.

The cost-planning process was developed progressively between the Expression of Interest (EOI) stage and the Early Contractor Involvement (ECI) stage. During Presentation 1, the purpose of the cost plan was to establish whether the proposed mixed-use Build-to-Rent development was financially credible at a conceptual level. The preliminary estimate therefore relied on schematic areas, broad elemental allowances and early assumptions regarding the live-rail interface. The EOI estimate established an indicative project cost of $280 million, based on an assumed gross floor area of approximately 23,000 m², an overall project rate of approximately $12,175/m², and an indicative range of $250 million to $320 million.

At the ECI stage, the cost-planning objective changed from establishing broad feasibility to improving cost transparency and traceability. The final Presentation 2 cost strategy therefore separated rail and temporary works, split building services into individual packages, introduced model-linked quantities, distinguished escalation from contingency, incorporated industry benchmarking and applied value-management principles.

This approach is consistent with the broader purpose of 5D BIM based cost management. BIM enabled quantity take-off can improve the speed and consistency of cost information while allowing design changes to be reflected more directly in quantities and estimates; however, professional judgement remains necessary for items that are not fully represented in the model, such as temporary works, risk allowances and site-specific construction constraints.

5.2 Development of the Cost Plan from EOI (Expression of Interest) to ECI (Early Contractor Involvement)

The original EOI estimate was intentionally broad because the project was still based on a schematic air-space development concept. The estimate considered the principal project characteristics, including a mixed-use residential building, more than 100 Build-to-Rent dwellings, retail and office uses, wellness facilities, an operational railway below the development and a major air-space transfer structure. The EOI cost plan therefore included relatively broad allowances for site establishment, rail interface works, foundations, the transfer platform, building structure, fit-out, services, approvals and uncertainty.

This approach was appropriate at the EOI stage because structural details, rail-interface requirements and detailed quantities had not yet been finalised. The original estimate was therefore explicitly identified as being subject to detailed quantity verification, structural development, SSD approval, TfNSW requirements, market testing and escalation.

During ECI, the estimate was developed into a more detailed elemental cost plan rather than simply replacing the $280 million figure with a new total. The purpose was to retain the $280 million as the current control estimate while providing greater visibility over the components that generated the cost. The process also demonstrated how the estimator's role interacted with the design, programme and construction methodology.

The final cost plan separated previously combined allowances. Rail-interface and temporary works were treated as distinct cost packages, building services were separated into mechanical, electrical, hydraulic/fire and lift packages, and the original combined contingency and escalation allowance was split into separate risk components. This improved transparency because the client could identify whether an allowance related to permanent construction, temporary methodology, future price movement or unresolved project risk.

Westmead Station Air-Space Mixed-Use Development

Indicative Estimated Cost: A$280,000,000
Assumed GFA: ~23,000m²
Overall Project Rate: ~A$12,175/m²
Indicative EOI Range: A$250m–A$320m

Element	Basis / Assumption	Indicative Rate / Basis	Cost (AUD)	%
Site establishment & preliminaries	Restricted site, cranage, management, live-station environment	Allowance	$15m	5%
Rail interface & enabling works	Rail protection, staging, temporary access, possessions/protection requirements	Allowance	$22m	8%
Foundations & substructure	Heavy foundations positioned around existing rail infrastructure	Allowance	$18m	6%
Air-space transfer platform	Major structural deck and transfer system above operational rail	~4,500 m² @ ~$8,440/m²	$38m	14%
Building superstructure	Podium + residential tower structure	~23,000 m² @ ~$1,390/m²	$32m	11%
Residential BTR fit-out	100+ essential-worker apartments	~14,500 m² @ ~$3,100/m²	$45m	16%
Retail, dining & offices	Café, restaurant, retail, serviced offices/co-working	~5,000 m² @ ~$3,400/m²	$17m	6%
Wellness & communal amenities	Gym, pool/spa, sauna, lounge and communal facilities	~1,500 m² @ ~$6,670/m²	$10m	4%
Façade & external works	Glazing, screens, masonry podium, terraces and landscaping	Allowance	$19m	7%
Building services & lifts	HVAC, electrical, hydraulic, fire, communications, lifts and pool plant	~23,000 m² @ ~$1,130/m²	$26m	9%
Design, SSD, TfNSW & BIM	Consultants, approvals, rail coordination and Digital Engineering	Allowance	$10m	4%
Contingency & escalation	Schematic-design uncertainty and future market movement	Combined allowance	$28m	10%
TOTAL			A$280m	100%

->

Westmead Station Air-Space Mixed-Use Development Itemised Cost Breakdown

Package	Cost (AUD)	%
Preliminaries + site establishment	$15m	5.4%
Rail interface + enabling	$14m	5.%
Temporary works / rail protection	$8m	2.9%
Foundations + substructure	$18m	6.4%
Air-space transfer platform	$38m	13.6%
Building superstructure	$32m	11.4%
Facade	$15m	5.4%
External works/ landscaping	$4m	1.4%
Residential BTR fit-out	$45m	16.1%
Retail/ office fit-out	$17m	6.1%
Wellness/ pool/ amenities	$10m	3.6%
Mechanical services	$10m	3.6%
Electrical / communications	$7m	2.5%
Hydraulic + fire services	$5m	1.8%
Lifts	$4m	1.4%
Design/ SSD/ TfNSW/ BIM	$10m	3.6%
Escalation allowance	$14m	5%
Contingency allowance	$14m	5%
TOTAL	A$280m	100%

Figure 5.2: Development of the cost plan from EOI to ECI (Source: ADDPI Developments, 2026)

5.3 Detailed Elemental Cost Plan

The final ECI control estimate remained at $280 million. The elemental breakdown was structured as follows:

Westmead Station Air-Space Mixed-Use Development
Itemised Cost Breakdown

Package	Cost (AUD)	%
Preliminaries + site establishment	$15m	5.4%
Rail interface + enabling	$14m	5.%
Temporary works / rail protection	$8m	2.9%
Foundations + substructure	$18m	6.4%
Air-space transfer platform	$38m	13.6%
Building superstructure	$32m	11.4%
Facade	$15m	5.4%
External works/ landscaping	$4m	1.4%
Residential BTR fit-out	$45m	16.1%
Retail/ office fit-out	$17m	6.1%
Wellness/ pool/ amenities	$10m	3.6%
Mechanical services	$10m	3.6%
Electrical / communications	$7m	2.5%
Hydraulic + fire services	$5m	1.8%
Lifts	$4m	1.4%
Design/ SSD/ TfNSW/ BIM	$10m	3.6%
Escalation allowance	$14m	5%
Contingency allowance	$14m	5%
TOTAL	**A$280m**	**100%**

Figure 5.3: Detailed elemental cost breakdown and major cost groups (Source: ADDPI Developments, 2026)

The breakdown demonstrates that the main financial exposure is not limited to the residential tower. Approximately $93 million, or approximately one-third of the control estimate, is associated with preliminaries and the live-rail/air-space interface, including rail enabling, temporary works, foundations and the transfer platform. The building shell and envelope account for approximately $51 million, while fit-out and amenity packages account for approximately $72 million. Building services and lifts contribute approximately $26 million, while design and approvals account for $10 million and escalation and contingency together account for $28 million.

The air-space transfer platform is therefore one of the most significant individual cost drivers at $38 million. Its cost reflects its structural role in allowing vertical development to occur above operational railway infrastructure while transferring loads to feasible structural support locations. Similarly, the $22 million combined rail-interface and temporary works allowance reflects the constraints associated with rail protection, restricted access, staging and construction beside operational transport infrastructure.

These allowances demonstrate that the project's cost per square metre cannot be compared directly with a conventional apartment building without recognising the infrastructure premium created by the station interface.

5.4 5D BIM Integration and Quantity Take-Off

The ECI estimate was supported by a 5D BIM cost-planning process. Presentation 2 developed the design from early Autodesk Forma concept analysis into an ArchiCAD-based developed building model. The costing workflow was then represented as:

BIM model □ Quantity take-off □ Unit rates □ Elemental cost plan □ Design/cost update

This process enabled key design elements to be connected with measurable quantities. The principal schematic quantities used included approximately:

- ➔ 23,000 m² total project GFA (Gross Floor Area)
- ➔ 14,500 m² of residential BTR (Build-to-Rent) area
- ➔ 5,000 m² of retail and office accommodation
- ➔ 1,500 m² of wellness and communal facilities
- ➔ 4,500 m² associated with the transfer platform basis

The intention of the 5D process was not to imply that every project cost could be generated automatically from the BIM model. Instead, the model provided a more traceable basis for measurable work, while the estimator retained responsibility for applying rates, allowances, construction methodology, temporary works, risk and escalation.

This distinction is important for a rail-interface project. Research into BIM-based quantity take-off notes that BIM can provide faster and more efficient quantity information, but some costs remain dependent on professional judgement where elements are not modelled or where project-specific methodology controls the final cost.

For example, a change to the transfer grid could alter structural steel and concrete quantities; a change in facade articulation could increase the number of panel types and installation complexity; and changes to apartment layouts could alter fit-out quantities and services distribution. Under the 5D approach, such design decisions could therefore be reviewed not only architecturally but also financially.

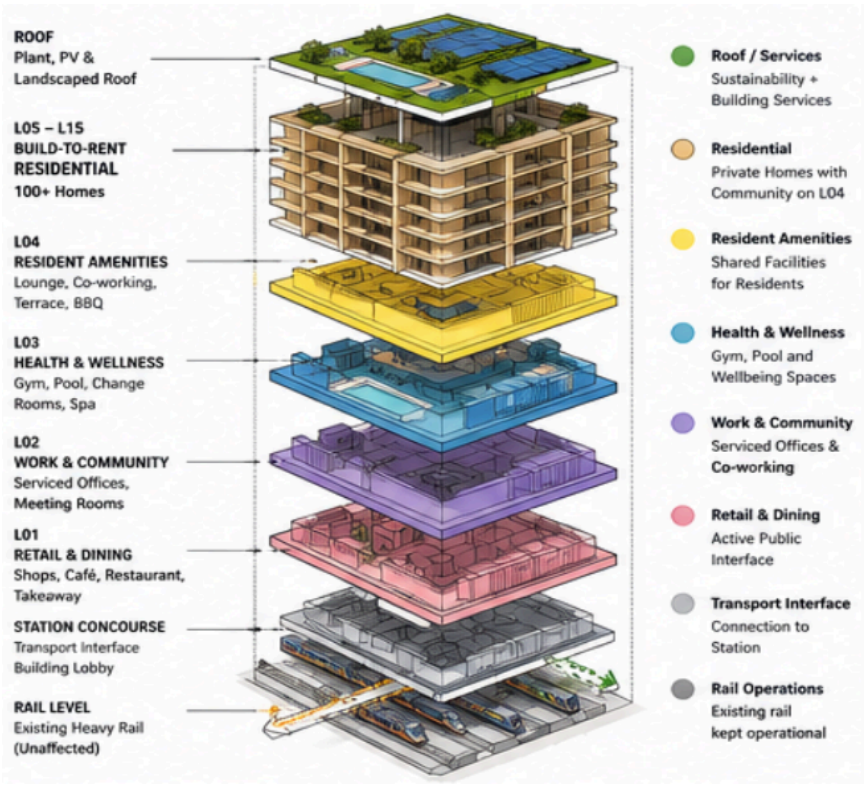


Figure 5.4: 5D BIM model-to-cost workflow (Source: ADDPI Developments,2026)

5.5 Industry Benchmarking and Financial Feasibility

Industry benchmarking was used as a reasonableness test rather than as a direct substitute for project-specific estimating. The submitted ECI analysis compared the Westmead project with Sydney residential benchmarks for multi-storey residential construction and identified typical 10-20 storey residential rates substantially below the project's overall rate of approximately $12,175/m².

RLB's 2026 Sydney construction-cost data places 10–20-storey residential construction in a materially lower range than the overall Westmead project rate, depending on apartment size and specification. This difference does not by itself indicate that the Westmead estimate is excessive, because the total Westmead rate includes costs that would not normally form part of a conventional residential benchmark.

The key project-specific premiums include the live-rail interface, air-space transfer structure, heavy foundations, temporary rail protection, restricted access, staging and specialist approvals. Therefore, the benchmark analysis was interpreted as:

> **Conventional building cost + rail-interface premium + transfer-platform premium + temporary works/restricted access = Westmead total project rate**

Crows Nest Over Station Development Site B was also used as a relevant reference project. The NSW Planning Portal describes the scheme as an over-station mixed-use and residential

development comprising 122 residential apartments, 78 car parking spaces and retail/commercial space within the station box, with the development approved on 16 April 2026.

Crows Nest was not treated as a direct cost comparison because differences in structural arrangement, scale, parking, procurement and rail conditions prevent a like-for-like financial comparison. Instead, it was used to demonstrate that residential and mixed-use development over major transport infrastructure is an established NSW development typology.

5.6 Value Management

Value management was used to improve the relationship between function, cost, constructability and programme rather than simply reduce scope. Research indicates that value management can improve project outcomes by considering cost, time and quality together, while BIM can strengthen value engineering decisions by providing improved design and cost information.

Five principal value-management opportunities were identified for the Westmead proposal.

First, the transfer platform was rationalised by seeking greater alignment between the tower/support grid and primary structural members. This reduces unnecessary transfer complexity and has the potential to reduce steel tonnage, bespoke fabrication and structural interfaces.

Second, the facade was reviewed with the objective of replacing excessive bespoke panel variation with more repeatable modular elements while retaining the intended architectural expression. This provides potential savings in fabrication, installation and future maintenance.

Third, apartment and wet area layouts were standardised where practical. Repetition of kitchens, bathrooms and vertically stacked service zones can reduce service runs, fit-out variation, waste and rework.

Fourth, mechanical electrical and plumbing (MEP) services were consolidated into coordinated risers and service zones through BIM. The objective was to reduce clashes, ceiling congestion and site based rework.

Finally, rail interface construction was reviewed for greater prefabrication and pre-assembly. Reducing the amount of work undertaken directly above or beside live rail can improve productivity while also reducing possession requirements and worker exposure.

Importantly, value management was not treated as uncontrolled scope reduction. Removal of required pool or wellness facilities was considered an inappropriate cost cutting measure because it would reduce the agreed project brief rather than improve the efficiency with which that function was delivered. The selected strategy therefore focused on maintaining project outcomes while improving the way they were designed, procured and constructed.

5.7 Temporary Works, Compliance and Escalation

The ECI cost plan separately recognised costs that could otherwise be underestimated in a conventional building estimate. These included temporary rail protection, hoarding, working

platforms, controlled access arrangements, construction staging and potential rail possessions. Compliance-related costs included SSD and TfNSW coordination, NCC/BCA requirements, fire performance, accessibility, acoustic and vibration treatment, specialist testing and digital engineering coordination.

Escalation was also separated from contingency because the two allowances respond to different risks. Escalation accounts for anticipated market price movement across the procurement and construction period, while contingency provides protection against unresolved design, quantity and interface uncertainty.

The project adopted an indicative $14 million escalation allowance and $14 million contingency allowance, maintaining the original combined $28 million risk provision.

This separation is justified by current market conditions. RLB's Q3 2026 Sydney market intelligence forecasts annual tender-price uplift of 4.0% in 2026, 4.5% in 2027, 5.0% in 2028 and 4.5% in 2029. As the project programme extends across these years, escalation exposure varies according to the timing of package procurement and expenditure. It would therefore be inappropriate to apply one undifferentiated percentage to the entire project value.

The $14 million figure remains an ECI level allowance rather than a final escalation calculation. At later design stages, escalation should be applied to package specific cash flows and procurement dates, particularly for long lead structural, facade, lift and building services packages.

5.8 Financial Impact of Design, Staging and Compliance Decisions

A key outcome of the ECI cost-planning process was recognition that financial management is not solely the responsibility of the estimator. Design, staging and compliance decisions all create measurable cost consequences.

From a design perspective, transfer-grid alignment affects steel and concrete quantities; facade articulation affects panel types and installation complexity; apartment standardisation influences repetition and waste; and the pool and wellness facilities require specialist plant and services.

From a staging perspective, the live rail environment increases requirements for preliminaries, rail protection, controlled cranage and restricted-access logistics. Rail-possession windows may also introduce temporary works and out-of-hours labour costs. In addition, the 36 month programme exposes the project to future escalation.

Compliance decisions also affect cost. SSD and TfNSW requirements create consultant and approval expenditure; NCC fire separation requirements affect specialist systems and testing; acoustic and vibration requirements influence isolation and facade performance; and BIM/Digital Engineering obligations create information management and coordination costs. These relationships were explicitly documented in the ECI financial literacy analysis.

This cross disciplinary cost awareness improves decision-making because design or programme changes can be assessed against their wider financial consequences before they are adopted.

5.9 Estimator Findings and Recommendations

The Estimator's ECI assessment concluded that $280 million should remain the project control estimate at the current level of design development, rather than being presented as a fixed tender price. The estimate has become more defensible because the cost plan is now separated into detailed work packages, linked to developed design quantities, benchmarked against external market information and supported by explicit allowances for temporary works, escalation and contingency.

The largest cost and uncertainty remain associated with the project's defining feature: construction above an operational railway. The transfer platform, rail-interface works, temporary protection, foundations and restricted access therefore require continued attention during technical design.

Further cost certainty should be developed through detailed structural take-off, verified BIM quantities, subcontractor and supplier market testing, package-specific escalation assessment and continued value-management workshops. In particular, the transfer structure, facade, services distribution and prefabricated rail-interface components should remain priority areas because they provide the strongest opportunities to improve cost efficiency without reducing required project outcomes.

Overall, the ECI cost-planning process demonstrates that financial feasibility is not determined solely by the headline $280 million figure. Feasibility depends on understanding how that figure is generated, what risks it contains, how design development changes it and where value can be improved while maintaining functionality, compliance, safety and project quality.

6 Project Manager (Alyta Huynh - 22062420)

6.1 Project Manager Role and Delivery Objectives

The Project Manager was responsible for coordinating the overall delivery strategy for the Westmead Station Air-Space Development. The role extended beyond preparation of the project programme and included coordination of design, approvals, procurement, cost, stakeholder requirements and construction planning across the project team.

The project presented a particularly complex management environment because the proposed mixed-use development was positioned above an operational heavy rail station. The delivery strategy therefore had to balance the progression of design and construction with continued railway operations, protected passenger movement, statutory approvals and the physical constraints of a restricted station site.

The Project Manager acted as the primary coordination point between the project's functional roles. During the EOI stage, the internal workflow placed the Project Manager at the centre of programme coordination, deadlines, meetings and information flow before information progressed through the Design Manager, Compliance Officer, Site Manager and Community Liaison Officer/Estimator. The team then used weekly coordination and Project Control Group meetings to review design, cost, programme and compliance impacts before finalising project decisions.

This approach reflected the Project Manager's principal objectives to: establish a realistic delivery programme, coordinate interdependent disciplines, identify programme constraints early, manage change and ensure that project decisions remained aligned with time, scope, cost and operational requirements.

6.2 Programme Development from EOI to ECI

The project programme was progressively developed between Presentation 1 and Presentation 2 rather than being treated as a fixed schedule from the outset.

At the EOI stage, the programme was structured around the RIBA Plan of Work and established the broad relationship between strategic definition, briefing, concept design, spatial coordination, technical design, construction, handover and use. TfNSW coordination was identified as an activity extending from the early design stages through construction, while station operations were to be maintained throughout the construction period.

The EOI programme also recognised three important relationships. SSD planning was intended to progress alongside design development, early procurement was to overlap with technical design, and station continuity required early TfNSW coordination with construction staged around rail and passenger operations.

Following ECI development, this broad strategy was refined into the final 36 month advanced master programme submitted in Presentation 2. The revised programme included more detailed

design, approvals, procurement and construction activities, together with programme milestones, controlled overlaps, critical path consideration and resource loading.

The reduction from the earlier broad programme to the final 36 month target did not rely solely on shortening individual activities. Instead, it was achieved primarily through improved sequencing and controlled concurrency. Design, statutory approvals and procurement were allowed to overlap where sufficient information was available, while construction and commissioning were progressively released rather than waiting for complete completion of every preceding work package.

This development from EOI to ECI demonstrates that the programme was treated as a live management tool rather than a static presentation output.

(INSERT IMAGE)

Figure 6.2: Development of the project programme from EOI to the final 36 month ECI master programme. (Source: ADDPI Developments, 2026)

6.3 RIBA Plan of Work Integration

The RIBA Plan of Work was adopted as the principal framework for structuring the project lifecycle. RIBA divides the development and operation of a building into eight stages from Strategic Definition through to Use and provides a framework for coordinating briefing, design, procurement, construction and handover activities.

For the Westmead project, RIBA Stage 0 established the strategic opportunity for air-space development above the station. Stage 1 addressed preparation, briefing, site investigations and early rail-interface constraints. Stage 2 developed the initial mixed-use and Build-to-Rent concept, while Stage 3 focused on spatial coordination between architecture, structure, services and the operational station.

Stage 4 developed the technical design sufficiently to release construction and procurement information. Stage 5 covered manufacturing and construction, including enabling works, rail protection, foundations, the air-space transfer platform and the residential superstructure. Stage 6 incorporated testing, commissioning, training and handover, while Stage 7 represented post-completion use and operation.

The RIBA framework was used as a management structure rather than as a strictly sequential programme. RIBA itself recognises that Stage 4 and Stage 5 commonly overlap within project programmes. This was particularly relevant to the Westmead strategy because selected construction and procurement activities needed to begin while later technical packages continued to be developed.

6.4 Final 36 Month Master Programme

The final ECI programme was structured around five major delivery periods.

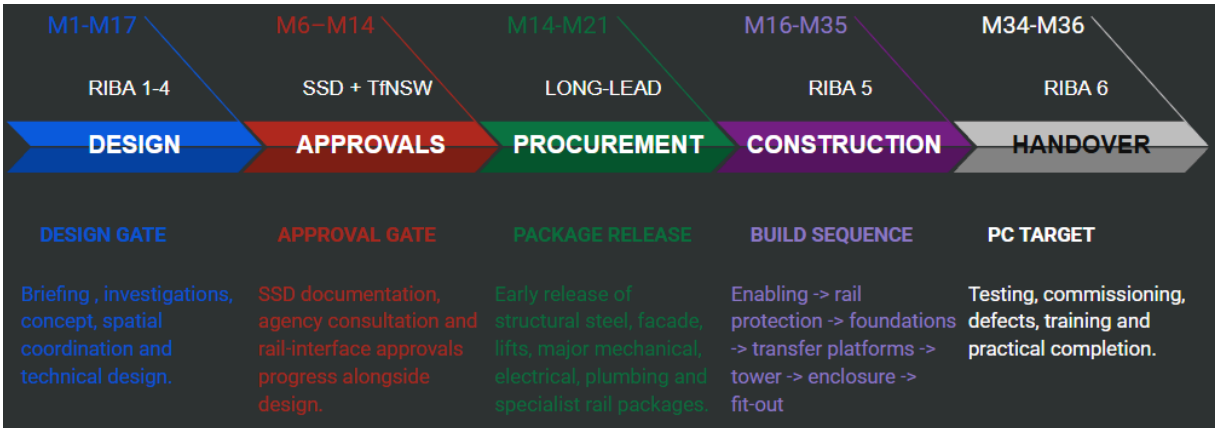


Figure 6.4: *(THINK OF CAPTION)*

The submitted sequencing strategy showed RIBA Stages 1-4 progressing through the design period, SSD and rail interface approvals overlapping the later design stages, early procurement beginning before construction, and RIBA Stage 5 commencing at approximately Month 16. Testing, commissioning and handover progressively occurred during Months 34 to 36.

Physical construction began with site establishment and enabling activities before progressing into rail protection, foundation works and the air-space transfer platform. The platform was followed by tower construction, enclosure, services and fit-out.

A major programme objective was to maintain Westmead Station operations throughout delivery. The original EOI strategy required the station to remain operational and identified station continuity as a continuing requirement during construction. The final ECI programme developed this further through protected passenger routes, rail protection, controlled access and rail-interface coordination.

The 36 month programme should therefore be interpreted as a target delivery programme rather than an unconditional guarantee. Its success is dependent on timely approvals, technical design development, rail access, procurement and the management of key interface risks.

6.5 Integrated Design, Approvals, Procurement and Construction Sequencing

One of the principal programme management decisions was to avoid an entirely sequential delivery model.

The first controlled overlap occurred between design and approvals. SSD documentation and agency consultation commenced while spatial coordination was still progressing. This reduced the risk of creating a lengthy approval period after design completion.

The second overlap occurred between design and procurement. Long lead packages could be progressively released once technical information had reached a sufficient level of development, rather than waiting for the entire technical design to be completed.

The third overlap occurred between procurement and construction. Early procurement was intended to support site mobilisation, rail-interface works and delivery of the transfer-platform package.

The final overlap occurred between construction and handover, where progressive commissioning could commence before the final fit out and close out activities were completely finished. These relationships were explicitly identified in the submitted ECI programme.

Controlled overlap shortened the overall programme but also increased coordination requirements. Information used for early procurement had to be sufficiently mature to avoid subsequent redesign or variation. The Project Manager therefore had to balance programme acceleration against design certainty.

6.6 Key Programme Gates

The final programme used a series of key gates to identify the points at which the project could safely progress to the next major phase.

The first major gate was SSD determination, representing the principal planning approval milestone. This was followed by rail protection readiness, confirming that the rail-interface and temporary-work arrangements were sufficiently coordinated for construction activities to progress.

The transfer platform completion milestone was particularly important because it represented the transition from complex rail-interface construction into the main vertical development above the station. The final gate was practical completion at Month 36, following testing, commissioning, defects management and training. The submitted programme identified these milestones visually to demonstrate the sequence of key delivery decisions.

These programme gates allowed the Project Manager to monitor progress against outcomes rather than monitoring duration alone.

6.7 Programme Constraints and Management Responses

Four principal constraints were incorporated into the programme: site access, weather, rail operations and construction staging.

Site access represented a significant constraint because the station environment provided limited laydown space and required construction movements to be separated from passengers. The programme response therefore relied on controlled deliveries, dedicated unloading and crane zones and segregated public and construction routes.

Weather primarily affected exposed activities such as crane operations, concrete, facade installation and external works. Weather allowance was incorporated into relevant packages, wind limits were to be monitored for lifting activities, and internal trades could be resequenced where external works were temporarily affected.

Rail operations constrained both the timing and physical location of construction activities. Ongoing TfNSW and Sydney Trains coordination, rail protection, exclusion zones and planned possessions were therefore required where normal operational separation was insufficient.

Construction staging was constrained by the structural dependency between the railway and the proposed development above. Vertical construction could not simply commence independently of the supporting platform. The programme therefore sequenced enabling works, foundations and transfer-platform construction before major tower works, followed by overlapping facade, services and fit-out activities.

These responses demonstrate that programme constraints were incorporated into the schedule rather than being managed only after construction commenced.

6.8 Indicative Critical Path Analysis

Critical-path consideration was used to identify the sequence of activities most likely to control the overall practical-completion date.

The submitted ECI critical-path diagram identified the principal sequence as:

> ***SSD determination rail / temporary-works design rail protection foundations transfer platform tower construction facade and MEP fit-out commissioning handover***

The transfer platform was identified as the principal programme gateway. Unlike a conventional tower project, vertical construction depends on completion of the structural system that transfers loads around the operational rail corridor. Delays to structural design, rail protection, foundations or the transfer platform could therefore propagate directly into the superstructure programme.

The critical-path assessment was described as indicative because the project remained at ECI stage and had not yet been converted into a fully logic-linked contractor baseline programme. At later stages, detailed predecessor relationships, activity float and subcontractor programmes would need to be confirmed through a scheduling platform such as Microsoft Project.

Nevertheless, the indicative analysis was valuable because it identified where programme monitoring and management attention should be concentrated.

6.9 Resource Management and Resource Loading

The ECI programme also incorporated an indicative workforce-loading assessment.

The resource profile demonstrated that project labour demand would not remain constant. During early design and approval periods, the project required relatively small numbers of design consultants, project-management personnel and rail-interface specialists. Resource demand increased as enabling, structural and transfer-platform works commenced.

The submitted resource plan reached an indicative peak of approximately 200 personnel at Month 33, when facade, mechanical, electrical, plumbing and internal fit-out activities were

operating concurrently. Major project resources also included tower and mobile cranes, piling and foundation plant, concrete pumps and specialist rail-protection, possession and safety crews.

Resource loading provided a check on the practicality of the programme. Excessive simultaneous activities could generate congestion, safety conflicts and inefficient use of site access, particularly within the restricted station environment. The workforce profile therefore supported decisions regarding staging, delivery planning, vertical transport and site supervision.

The workforce assumptions remain indicative at ECI stage and would require refinement following subcontractor engagement and detailed construction planning.

6.10 Procurement and Long-Lead Package Strategy

Procurement was integrated with the programme because several packages could create significant delays if ordered only after all technical design had been completed.

The ECI sequencing identified an early package-release period from approximately Month 14 to Month 21, covering structural steel, facade, lifts, major mechanical, electrical and plumbing systems and specialist rail packages.

Early release of these packages provided several programme benefits. It allowed manufacturing lead times to run concurrently with later design and enabling works, supported construction readiness and reduced exposure to delays from specialist materials and equipment.

However, early procurement also introduces design-freeze risk. A package released before sufficient coordination can create variation, rework or interface conflicts. For this reason, package release needed to be tied to defined design maturity and approval gates.

The broader NSW Government delivery environment also supported a collaborative contract-management approach. GC21 Edition 2 is supported by contract-management tools, role descriptions, meetings and workshop guidance, making structured communication and risk management relevant to a project of this complexity.

6.11 Project Risk Management

Programme risk management focused on those events most capable of affecting the 36 month delivery target.

The primary risks included delay to SSD determination, unresolved TfNSW interface requirements, rail possession availability, transfer-platform design development, long-lead procurement, restricted site access, adverse weather and escalation related pressure on package procurement.

The management strategy relied on early identification and integration rather than maintaining a separate risk process. SSD documentation progressed alongside design; TfNSW coordination commenced early; rail interface design preceded physical rail works; selected procurement

occurred before construction; and weather-sensitive activities were supported by alternative internal work fronts.

Risk management was therefore closely linked to the programme and cost plan. For example, a delayed rail interface approval could extend preliminaries and escalate procurement costs, while late transfer platform design could delay the critical path and affect the release of superstructure resources.

The final technical report should therefore treat programme, cost and construction risk as interdependent rather than independent project controls.

6.12 Digital Collaboration and Project Controls

Digital collaboration was a central component of the Project Manager's internal workflow.

During the EOI stage, the team documented a shared workflow using Autodesk Forma Board and a shared drive, supported by weekly coordination and PCG meetings. Microsoft Excel was used for preliminary programme development, Google Drive, Docs and Slides supported shared information and collaborative editing, while WhatsApp was used for day-to-day communication and organisation of meetings. Autodesk Forma Board was used to record project tasks, milestones, progress updates and coordination activities.

This distinction between informal and formal communication was important. Messaging platforms enabled rapid communication, whereas the project board and shared document environment provided a more structured record of project actions and information.

The project also adopted a digital engineering approach consistent with the broader TfNSW requirement for coordinated project information. TfNSW's Digital Engineering Framework establishes an agency level framework for digital engineering and information management across transport projects.

For the Project Manager, digital collaboration provided visibility across the design, programme, cost and compliance workstreams and allowed decisions to be reviewed across multiple disciplines before being incorporated into the project.

6.13 Change Control and Cross-Discipline Coordination

An important lesson from the project was that changes to one discipline could not be considered in isolation.

For example, a modification to the transfer platform could affect structural quantities, the cost plan, technical design duration, rail-interface approvals, temporary works and construction sequencing. Similarly, changes to facade design could affect procurement lead times, cost, crane requirements and installation sequencing.

The project team's workflow therefore operated as a coordinated feedback loop:

Design change □ coordination review □ cost impact □ programme impact □ compliance review □ site/logistics review □ coordinated decision.

Weekly PCG meetings provided the mechanism for reviewing these interactions and assigning follow up actions. The workflow established in Presentation 1 specifically required design, cost, programme and compliance to be updated following coordinated review.

This process improved information flow and reduced the risk that one discipline would progress on assumptions that had not been accepted by the wider project team.

6.14 Project Management Findings and Recommendations

The final ECI planning exercise demonstrated that the Westmead proposal could be structured around a 36 month target programme, provided that design, approvals, procurement and construction were actively coordinated rather than managed as independent sequential phases. The strongest programme opportunities arose from controlled overlap between design and approvals, early release of long-lead packages and progressive commissioning. Conversely, the greatest programme risks remained associated with rail interface coordination, statutory approvals and the transfer platform.

The transfer platform should continue to receive priority management because it represents the principal interface between the operational railway and the vertical development and forms a major gateway on the indicative critical path.

Resource-loading analysis also demonstrated that the later construction period would require careful control of workforce congestion, deliveries and vertical logistics as facade, MEP and fit-out activities overlap. The indicative peak of approximately 200 workers should therefore be refined through detailed subcontractor programmes before construction.

Digital collaboration should continue through a controlled common information environment supported by formal task tracking, programme updates and regular PCG meetings. This is particularly important because early procurement and overlapping activities increase the consequences of late information or uncoordinated change.

Overall, the Project Manager's role evolved from producing a broad EOI timeline into establishing an integrated delivery control strategy. The final programme linked RIBA stages, SSD and rail approvals, procurement, critical path activities, resources, constraints and multidisciplinary coordination into a single delivery framework. This progression demonstrates that effective project management is not limited to identifying when activities occur; it requires continuous management of the dependencies that allow the design, approvals, cost plan and construction methodology to progress together.

(ADD FIGURES/ EVIDENCE)

7 Community Liaison Officer (David Azmy - 20526304)

7.1 Regulatory Compliance and Statutory Approval Framework

The proposed westmead station re-use and mixed used development is a larger and more complex project that requires careful planning, approvals and compliance with building regulations. The project aims to improve the existing station by introducing residential Apartments, commercial spaces, community facilities and better transport connections. As the development is in close proximity to a liver operating railway infrastructure. These challenges must be identified early and managed through the project lifecycle to ensure that construction works are carried out for safety and efficiency and no distribution to general operation of the railway for commuters.

As outlined in the ADDPI expression of interest and early contractor involvement the project needs to meet multiple important requirements. This is shown by obtaining necessary planning approvals following the national construction code and coordinating with transport NSW (TFNSW), applying safety in design principles and the use of digital engineering to support project coordination (ADPPI developments 2026a, 2026B). These requirements are in place to ensure the project is compliant to meet regulations to reduce disruptions to railway operations and surrounding infrastructure. The compliance officer plays an important role in ensuring the project follows all relevant laws and regulations and approvals as well. This includes the processing of necessary approvals and document coordination, monitoring throughout the project with relevant authorities. Having a clear compliant management process will help to ensure that the design, procurement and construction stages are met and the consideration of challenges of working near a live and active railway are met.

7.2 Statutory Approvals and state significant development pathway.

The proposed Westmead station reuse development requires multiple planning approvals to ensure that the project is adhering to the NSW planning laws and also the railway requirements that are in place. It is known that the development is located in a sensitive location that is close to an operating live railway station where commuters are travelling. Additional consideration must be considered when compared to a regular mixed use development. These include railway protection to the existing infrastructure, railway safety, public access and managing environmental impacts.

Under the environment planning and assessment Act 1979 (NSW) developments must obtain the necessary planning approvals before construction can begin. Depending on the size and type of the development, the project may be classified as a state significant development commonly known as (SSD). However, further to that assessment would be required to determine and examine if the westmead project meets the requirement approval process. As mentioned in the ECI presentation the project will involve early communication with TFNSW and

planning authorities and council and any other relevant authorities stakeholders (ADDPI developments 206B). This will ultimately help to identify the necessary approvals during the design stage and reduce the risk of delays or outstanding compliance requirements.

If the project is classified as a SSD, then the approval process would then generally involve a preparation of an environmental impact statement which assesses the impacts that will occur on the environment and surrounding suburbs and community and obtaining an approval from the relevant authority. As for the westmead development an impact factor to consider is the construction noise that will be generated from construction and vibration, pedestrian access and traffic impacts on the existing railway infrastructure. The compliance officer will be responsible for maintaining an approval register that records the required approval that needs to be obtained from the relevant authorities, responsible personnel and the status of each submission. The register will help the project team keep track of outstanding approvals and improve communication between the design team, project managers and external authorities. It will also help to ensure that all the necessary approvals are obtained before the relevant construction activities are undertaken and begin.

7.3 National Construction Code and Building Code of Australia Compliance.

The proposed development in Westmead station reuse and mixed use development must follow the National construction code which sets out the minimum standards for building safety, accessibility, structural requirements and fire protection to the building (ABCB 2022). As the project includes residential apartments, commercial spaces and community facilities each part of the developments must be classified correctly to ensure that it meets the relevant building measures and requirements.

Fire safety is crucial and an important part of the project especially in recent years laws and regulation have changed to add extra protection in fire requirements in a building, and particularly in this project as it is close to an operating railway. The building design must include suitable fire separation, emergency evacuation routes and egress pathways and also smoke management system to be in place all which will be mentioned in a fire engineering report. The development must also meet the accessibility requirements outlined in the NCC and relevant standards including AS 1428.12021.

The compliance officer will work closely with architects, Fire engineers and building certifications to review building plans and documentation, to help identify and not compliance or standards that are not met or achieved and to ensure these issues are addressed accordingly before the construction begins.The will help to ensure that building safety, reduced risk of compliance issues and avoidance of any unnecessary design changes and construction delays.

7.4 GC21 Contractual Compliance

The proposed westmead station re-use and mixed development requires a suitable construction contract to clearly outline the responsibilities of everyone involved, manage project risks that may occur during the construction stage and to ensure that all legal requirements and stansrds

are followed correctly. The NSW Government GC21 General conditions of contract provides a framework that encourages teamwork, clear communication, fair allocation of risks and resolving disputes between the parties involved in construction.

As the project is located near an operating railway, the contract must also consider railway access restrictions, construction safety, protection of existing infrastructure and coordination with transport for NSW (TFNSW). These requirements are important to ensure construction activities do not significantly affect the railway day to day operations and to also help avoid risk for the project for delays. The compliance officer will be on top of the contract requirements, maintain records of planning approval and will ensure that all compliance responsibilities are clearly communicated to the project team. This will include managing all documents, identifying potential compliance risks and checking that the project continues to meet regulatory requirements throughout construction. Using the GC21 contract principles can help improve communication, accountability and coordination between all stakeholders. This will support delivery of the westmead development while ensuring the project follows all relevant laws and regulations.

7.5 Transport for NSW (TFNSW) Digital Engineering Framework

The proposed development requires the use of digital engineering tools to help in management and coordination of construction activities near the operating railway. The transport NSW (TFNSW) digital engineering framework provides guidelines for managing project information and improving coordination between the design team to ensure that the project meets the required standards throughout construction (TFNSW 2022).

As outlined in ADDPI developments ECI presentation, Building information modelling will be used to support the design and coordination of the project (ADDPI Developments 2026B). Software such as Autodesk and revolt can help to identify possible clashes that will happen with the building services and also identify potential issues with the existing railway infrastructure that may occur.

The compliance officer will work closely with the design and engineering teams to review digital drawings and documentation, identify any issues that relate to building regulation or compliance issues and ensure everything is designed up to Australian standards and adhering to the NCC. This will inevitably help to identify potential risks early on and reduce the expenses of design changes during the construction phase of the project. The use of digital engineering throughout the project team will significantly improve the management of the project information throughout all including stakeholders and monitoring compliance requirements. This will also help to ensure that construction activities are carried out safely and efficiently near the operation of live railways.

7.6 Safety in Design and Railway Compliance

The proposed development requires a comprehensive plan for safety to reduce construction risk, especially as the project is located in close proximity to a live operating railway. This increases the risk of safety to the public. Under the work health and safety act 2011 NSW (NSW) everyone in the construction site is liable and responsible for their own safety, not the site managers or project manager but all those on site are responsible for their safety. This includes identifying potential hazards and putting suitable safety measures in place during the design and construction stages. As mentioned in the ADDPI developments ECI presentation the construction strategy considers railway operations, pedestrian safety, limited site access and possible impacts on existing infrastructure (ADDPI developments 2026b). These challenges require close coordination with transport NSW and other railway authorities to ensure that construction activities do not intervene or affect passenger safety or to affect railway services lines.

Some of the main safety risks include construction vibration, unauthorised access to the railway areas, movement of construction machinery and to also maintain safe pedestrian pathways. The way to minimise the risk of the project can use staged construction, designated exclusion zones and controlled site access to ensure work is carried out safely as mentioned in the presentation. The compliance officer will work closely with design teams, safety officers and railways authorities to review the safety design documentation, monitor safety regulations and identify possible compliance issues that arise before the construction phase of the project. This will include reviewing and assessing subcontractors safe work methods statements commonly known as (SWMS) and safety management plans form the head contractor that are to be in place during the construction of the developments. By ensuring safety in the design of the project is incorporated and adhered to it will help to improve coordination and reduce disruptions to the railway operations. This will also help ensure the development meets the relevant NSW workplace health and safety requirements.

7.7 Compliance Matrix and Monitoring Strategy

The Proposed Westmead station reuse and mixed development requires a clear compliance monitoring management plans to ensure that all legal and contractual obligations are meet along with railway safety required throughout the project. As outlined in the ADDPI developments ECI presentation, the compliance process includes obtaining approvals, coordinating design form consultants and engineers and managing railway activities and regularly checking that project requirements are being followed to standards. A compliance matrix has been prepared by responsible parties and supporting documents. This would help ideally for the project team to identify compliance issues that may arise during the design stage and construction stages. This will ultimately help to reduce the risk of delays, design changes to occur and additional cost added to the project.

Proposed Compliance Matrix

COMPLIANCE AREA	APPLICABLE REQUIREMENTS	RESPONSIBLE ROLE	PROPOSED COMPLIANCE EVIDENCE
Statutory planning	EP&A Act 1979 and applicable planning approval pathway	Compliance Officer / Planning Consultant	Planning assessment and approval documentation
NCC/BCA	NCC and relevant Australian Standards	Architect / Building Certifier	NCC compliance assessment and design certificates
Railway interface	TfNSW and relevant rail authority requirements	Compliance Officer / Rail Interface Manager	Interface agreements, access approvals and rail safety documentation
Safety in Design	WHS Act 2011 and WHS Regulation 2017	Design Manager / WHS Coordinator	Safety in Design register and risk assessments
Digital engineering	Applicable TfNSW Digital Engineering Framework requirements	BIM Manager / Design Manager	Coordinated BIM models and design review records
Environmental management	Applicable planning consent and environmental requirements	Environmental Manager	Environmental management plans and monitoring records

The compliance officer would generally regularly update the compliance matrix during the design and construction stages as new documents and information becomes available for the project from consultants and engineers. Regular meetings also are held with the project manager, design consultant and relevant authorities to identify if there are any missing approvals or and noncompliance issues. As mentioned in the ECI presentation digital engineering and communication between stakeholders parts of the project (ADDPI developments 2026b). A central document management sharing system such as a share drive or dropbox would be used to keep track of design changes, inspections or any compliance reports.

Special attention would be given to the railway requirements, that also including construction access, passenger safety and protecting the existing railway infrastructure. Any construction work affecting railway operations would need the required approvals and safety measures in

place before starting. If any noncompliance issues are raised the required actions would then be noted and recorded and then assigned to the responsible team members. These issues would then be monitored until they are properly resolved and additional supporting evidence will need to be provided. This will ultimately reduce compliance risks, improve accountability by each individual responsible and will also ensure that requirements are followed throughout the project rather than in comparison to only being checked at the final certification stage.

8. Site Manager (Philip Semenov)

Responsible for logistics, construction safety, and on-ground constructability, ensuring the project can be delivered efficiently and safely, especially in the context of a live railway environment.

8.1 : Site managers role and scope

The Site Manager is accountable for logistics, construction safety and on-ground constructability in relation to the Westmead Station Air-Space Mixed-Use Development. The problem statement is that a 16-storey Build-to-Rent (BTR) structure, containing over 100 apartments for essential workers, will have to be erected above an active heavy rail station. The Westmead Station is meant to stay operational throughout all 36 months of the advanced master programme (ADDPI Developments 2026b).

This section outlines the evolution of ADDPI's construction method from the Expression of Interest (EOI) stage to the Early Contractor Involvement (ECI) stage. During EOI, the Site Manager outlined three key constraints – pedestrian safety, active rail system and structural interface with tracks and platforms (ADDPI Developments 2026). During ECI, these constraints were translated into a staged construction, temporary works and rail protection method, cranage strategy, modular construction, and the Safety in Design (SiD) process. Each of these points is examined below with reference to figures in the presentation and the programme.
The project brief determines the scope of work in terms of three duties: logistics, construction safety and on-ground constructability. These are assessed with regard to two outputs – efficient delivery and safe delivery in a live rail environment. This is shown in Table 8.0 below

Responsibility	How it is addressed	Sections
Logistics	Segregation of access, JIT deliveries, deck lay-down, sizing of workforce/worksites	7.2, 7.5
Construction safety	Railway exposure grades, possessions, temporary works, SiD register	7.4, 7.5, 7.7
On-ground constructability	Construction sequencing, transfer platform staging, pre-fabrication, hold points	7.3, 7.6
Live railway environment (all three)	Station remained open for M1 to M36; DZ work restricted to possessions	7.4

Table 8.1 : Site managers responsibilities and the way they are managed

The Site Manager's decision-making does not stand alone; it is responsible for the following cost centers in the ECI cost estimate:

- Preliminaries and site setup ($15 million)
- Rail interface and enabling works ($14 million)
- Temporary works and rail protection ($8 million)
- Foundations and substructure ($18 million)
- The air space transfer platform ($38 million)

In total, these amount to $93 million (33%) of the $280 million control estimate (ADDPI Developments 2026b).

8.2 Planning of logistics and Site constraints

The area size is 10,170 m² according to a boundary survey in Autodesk Forma with dimensions of 51.45 m x 181.54 m x 64.48 m x 172.92 m. The project site lies within the SP2 Railway corridor. Hawkesbury Road is on the western side, Railway Parade on the northern side and Alexandra Avenue on the southern side. The vicinity also consists of R4 High Density Residential zoning, schools and the Westmead hospital precinct (ADDPI Developments 2026). The Forma topographic survey indicated that there is a fall towards the east-south-east on the site. This impacts excavation depths, drainage during construction and working platform elevations

The site access was established as the first programme constraint in the ECI programme because of restricted station site, limited laydown areas and interaction with commuters (ADDPI Developments 2026). The site footprint is dominated by tracks and platforms and therefore conventional laydown areas are not available. The logistics plan deals with this constraint through five strategies.



Figure 8.2 – site zoning and area (Source: ADDPI Developments 2026)

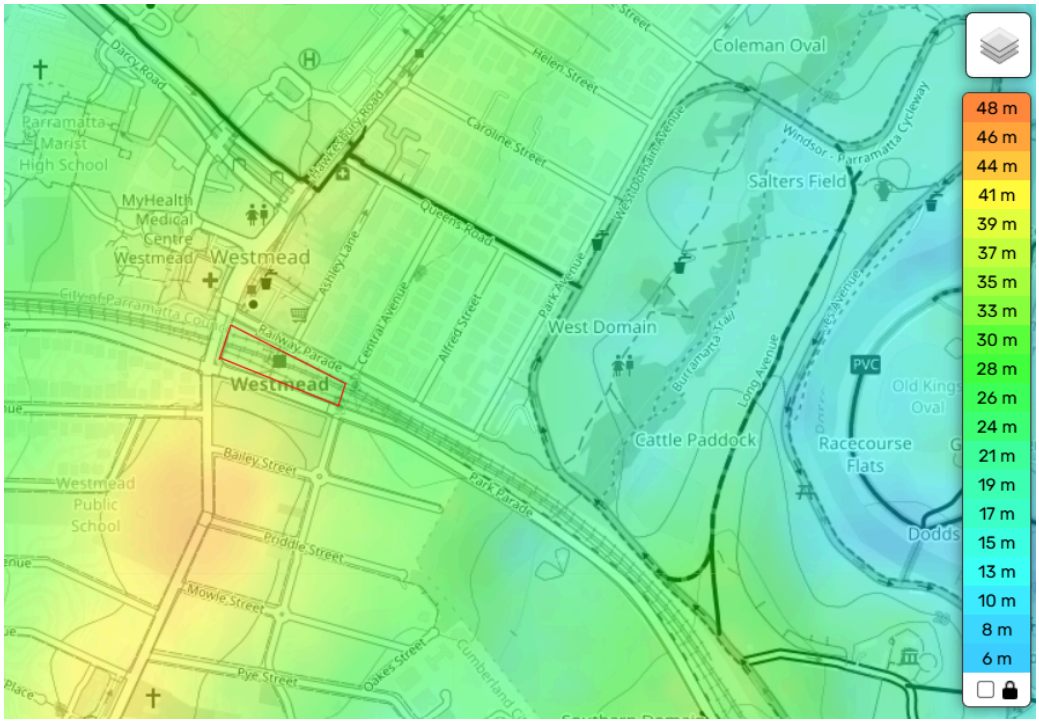


Figure 8.3 – topography and elevation study (ADDPI Developments 2026)

1. Segregated routes. The construction access is positioned off Railway Parade and Alexandra Avenue from the station entrance from Hawkesbury Road. Pedestrian routes are hoarded so that public circulation remains physically segregated from the work zone, which is consistent with the EOI circulation strategy (ADDPI Developments 2026).

2. Just-in-time deliveries. Materials are ordered as per each lift sequence as opposed to stocking and deliveries are scheduled at times when they do not coincide with commuter peaks in the morning and evening periods.

3. Unloading and crane zones. The loading bays are positioned inside the hoarding line so that trucks unload materials within the hoarded area as opposed to public footpaths.

4. Elevated laydown. Once the transfer platform is completed in Month 26, the deck forms the main laydown and staging area for the superstructure. It reduces the demands on surrounding streets during the peak construction period.

5. Off-site consolidation. Façade panels, bathroom and kitchen pods and transfer platform components are pre-assembled in an off-site consolidation yard and delivered in lift order. It can be seen from the resource histogram in the ECI presentation that peak demand is expected (Figure 8.4).

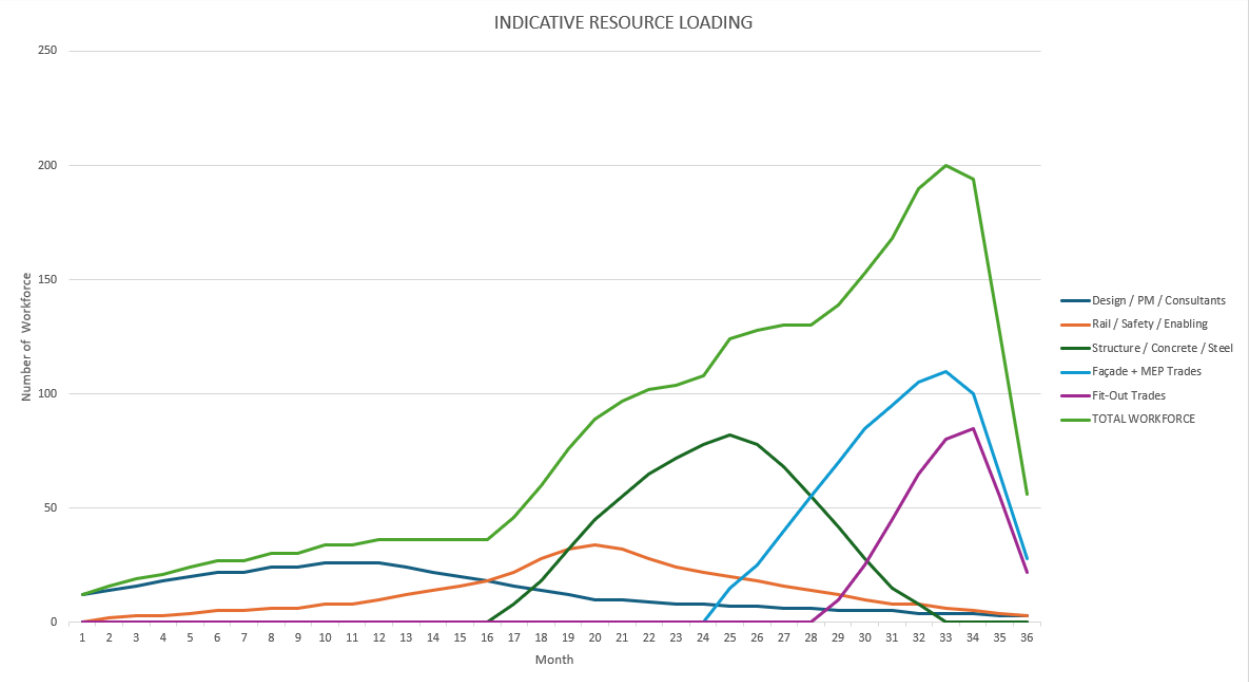


Figure 8.4 – Indicative resource loading histogram, 36-month programme (ADDPI Developments 2026)

The workforce will increase from 36 workers in Month 16 to a peak of 200 in Month 33 (ADDPI Developments 2026b). At this stage, all façade, MEP and fit-out trades are operating simultaneously. Therefore amenities, vertical transport and material hoists are designed for 200 workers. The hoists are located close to the tower core so that workers' movement does not interfere with public station routes.

8.3 Construction Methodology and Staging

There is a rigid bottom-up dependency in construction: anything cannot go up until there is an enclosed deck that can bear loads. ECI programme provides a sequence of enabling works, rail protection, foundations, transfer platform, tower, enclosure, and fit-out (ADDPI Developments 2026).

Table 8.2 – Construction Staging (Based on ADDPI Developments 2026)

Stage	Months	Site management focus
1. Site establishment and hoarding	M16–M18	Delineate works from commuters before heavy works start
2. Utility protection and enabling	M17–M19	Prevent strikes to rail signalling, power, and communication
3. Rail protection and temporary works	M18–M20	"Milestone: rail protection ready" before working above rails

4. Foundations and substructure	M19–M23	Piling above railway only during possession period
5. Air-space transfer platform	M22–M26	Critical path activity; segments installed under possession periods
6. Superstructure	M25–M30	Cranage, wind restrictions, and edge protection
7. Façade, MEP and fit-out	M27–M34	200 workers peak at M33
8. Station interface and commissioning	M32–M36	Restoration of permanent public routes

Transfer Platform construction is what the Site Manager cannot afford to miss. All further packages are dependent on it, and this can only be constructed in possessions allocated in the Sydney Trains timetable. There are two ways in which this can be safeguarded. The structural steel is sourced in Month 14 so that segments are prefabricated prior to the first possession, and an extra possession is allocated for every critical lift to safeguard against any cancellation impacting Month 26.

8.4 Live Rail Management

Activities adjacent to the corridor shall conform to Rail Safety National Law (NSW) and access and protection provisions of Sydney Trains (Rail Safety National Law (NSW) No. 82a 2012). Provisions for the management of rail corridors of the State Environmental Planning Policy (Transport and Infrastructure) 2021 will require approval from the rail authority for any adjacent excavation works. This means that ADDPI classifies site activities based on rail exposure as follows:

• Tier 1 – outside the corridor: site establishment, off-corridor piling and fit-out above the deck, under normal site control procedures.
• Tier 2 – adjacent to the corridor: protection screens, exclusion zones and rail protection officers; proceeds under train services.
• Tier 3 – inside or above the danger zone: piling between tracks, girder installation and lifting over tracks, only under track possessions.

It translates the EOI commitment to "keep rail moving" to an easily verifiable on-site policy. Each possession has a work plan including lifting sequence, crew numbers, hand-back time and a decision point, when unfinished work becomes safe instead of continuing after possession end. Driven bored piles are used instead of drilled, subject to assessment of vibration levels according to the guidelines: Assessing vibration: a technical guideline (Department of Environment and Conservation NSW 2006). Overhead protection decks are provided for platform protection, and crane slewing is not allowed over occupied spaces.

8.5 Temporary Works and Crane Study

At ECI stage, the separate contract for temporary works and rail protection ($8m) has been developed in distinction to rail interface and enabling works ($14m) to be designed, costed and approved as its own package (ADDPI Developments 2026). The main temporary works are protection screens, overhead protection decks, shoring for transfer segments and crane platforms on sloping ground. Each of these is designed by a temporary works engineer, independent assessment and, where applicable, review by Sydney Trains during Months 12–18 design activity.

The crane plan includes selection of each crane for its lifting tasks. The heaviest loads are transfer segments lifted with a large mobile or crawler crane erected in the off-corridor area, to allow clearance in the corridor between possessions. Shapira, Lucko and Schexnayder (2007) point out that selection of cranes on constrained sites is dictated by the heaviest critical lifts. After completion of the deck, a tower crane on the podium lifts superstructure, with slew limiting to avoid loads over occupied platforms and live wiring. All crane lifts comply with AS 2550.1-2011 (Standards Australia 2011) and wind speed limitations.

8.6 On-Site Constructability and Modular Construction

The ECI proposal was a sequential four-module process involving enabling works and rail protection, foundation with rail crew, possession lift prefabricated transfer units, and prefabricated modules above the deck (ADDPI Developments 2026b; Figure 7.5). Value management replaces site fabricated structures above live rail with prefabricated rail interface, thus reducing possession time and risk of safety issues. According to Gibb (1999), offsite fabrication means relocation of construction work from hazardous and restrictive site conditions to more favorable locations.

The proposed lightweight hybrid structure also supports this: steel transfer slab, cross-laminated timber/ glue-laminated wood floors, and lower carbon concrete podium and cores. The lighter structure results in reduced loading on the transfer platform and smaller possession lifts. The repetitive residential floor plates between Level 4 and Level 15 allow for the standard floor cassettes with prefabricated bathroom and kitchen pods (Figure 7.6). ADDPI suggests this

method rather than full volumetric apartments due to additional demands on tower crane capacity.

On-site confirmation of constructability includes hold points: pile positions checked against the transfer grid, the first lift of the transfer segment, and the mockup of one residential bay.

8.7 Construction Safety: Safety in Design and WHS

Designers should ensure that any hazard for the people involved in the building, use and maintenance of the structure is minimized under the Work Health and Safety Act 2011 (NSW). According to Safe Work Australia (2018), it is advised to control the hazards at the design stage where controls have the highest priority in the hierarchy of control. In that regard, the Site Manager organizes the SiD workshop at RIBA Stages 2 to 4 as captured in the Compliance Matrix (ADDPI Developments 2026). Hazards and the control measures are highlighted below with the priority to design stage controls and the secondary being the site controls (addition to ADDPI Developments 2026a, 2026).

• Workers struck by trains or electrocution: Prefabricated transfer units minimize working on track, and all the Tier 3 work is done under possessions. At the site, there will be rail protection officers and overhead wiring isolation.

• Falling objects on commuters: Overhead protection deck and crane slew limits will be used, along with exclusion zone and tool tethering.

• Commuters entering into work zones: Public and construction routes will be segregated in the logistics design, and hoarding and traffic controllers will enforce the segregation.

• Collapse during the erection of the transfer platform: Transfer grid and engineered temporary propping will minimize the hazard, independent checks and hold points before each release of stages.

• Vibration damage to the station: Bored piles and light superstructure minimization of vibrations at the source, and real-time monitoring and trigger levels for stopping the process if the limit is reached.

The hazard register provides input for the WHS management plan and safe work method statements as required by the Work Health and Safety Regulation 2017 (NSW) and Safe Work Australia (2018a).

8.8 Evidence of Digital Engineering

There were two pieces of technology that were used by the Site Manager in his planning process. One technology that was used at EOI level was Autodesk Forma which involved feasibility study of the site involving boundary/area study, topographic study and study of seasonal sun hours (ADDPI Developments 2026). It provided the logistics constraints mentioned in section 7.2, such as working platform levels on the sloped land. ArchiCAD created a model from the design that was carried out at the ECI level and this involved zoning of the commercial podium and the residential levels above it (ADDPI Developments 2026). The recurring floor plates of residential floors form the basis of cassettes and pods in section 7.6.

9 Conclusions

Briefly summarize the key points

10 References

As necessitated, all quality sources appropriate for the project.
Referencing uses the WSU Harvard system (see the link below). Marks will be deducted for improper referencing.

ADDPI Developments 2026a, *Expression of interest: station re-use – Westmead Station*, group presentation, Complex Building Project, Western Sydney University, Sydney, 18 August.

ADDPI Developments 2026b, *ECI: early contractor involvement – station re-use: Westmead Station*, group presentation, Complex Building Project, Western Sydney University, Sydney, 29 September.

Department of Environment and Climate Change NSW 2009, *Interim construction noise guideline*, DECC NSW, Sydney, https://www.epa.nsw.gov.au/sites/default/files/09265cng.pdf.

Department of Environment and Conservation NSW 2006, *Assessing vibration: a technical guideline*, DEC NSW, Sydney, https://www.epa.nsw.gov.au/sites/default/files/vibrationguide0643.pdf.

Gibb, AGF 1999, *Off-site fabrication: prefabrication, pre-assembly and modularisation*, Whittles Publishing, Caithness, Scotland.

Kineber, AF, Othman, I, Oke, AE, Chileshe, N & Buniya, MK 2021, 'Impact of value management on building projects success: structural equation modeling approach', *Journal of Construction Engineering and Management*, vol. 147, no. 4, https://ascelibrary.org/doi/10.1061/%28ASCE%29CO.1943-7862.0002026.

Li, X, Wang, C & Alashwal, A 2021, 'Case study on BIM and value engineering integration for construction cost control', *Advances in Civil Engineering*, vol. 2021, article 8849303, https://onlinelibrary.wiley.com/doi/10.1155/2021/8849303.

NSW Department of Planning, Housing and Infrastructure 2026, 'Crows Nest OSD Site B', SSD-61400212, NSW Planning Portal, https://www.planningportal.nsw.gov.au/major-projects/projects/crows-nest-osd-site-b.

Rail Safety National Law (NSW) No. 82a 2012, https://legislation.nsw.gov.au/view/html/inforce/current/act-2012-82a.

Rider Levett Bucknall 2026a, 'Australia market intelligence Q3 2026: Sydney', RLB, https://www.rlb.com/insights/oceania/australia-market-intelligence-q3-2026/sydney.

Rider Levett Bucknall 2026b, *Riders digest Sydney 2026*, RLB, https://www.rlb.com/wp-content/uploads/sites/1/2026/01/2026-RLB-Rider-Digest_Sydney_Digital_2.pdf.

Royal Institute of British Architects 2020, *RIBA plan of work 2020*, RIBA, London, https://www.riba.org/work/insights-and-resources/riba-plan-of-work.

Safe Work Australia 2024a, *Model code of practice: construction work*, Safe Work Australia, Canberra, https://safeworkaustralia.gov.au/sites/default/files/2024-11/model_code_of_practice-construction_work-nov24.pdf.

Safe Work Australia 2024b, *Model code of practice: safe design of structures*, Safe Work Australia, Canberra, https://www.safeworkaustralia.gov.au/sites/default/files/2024-11/model_code_of_practice-safe_design_of_structures-nov24.pdf.

Shapira, A, Lucko, G & Schexnayder, CJ 2007, 'Cranes for building construction projects', *Journal of Construction Engineering and Management*, vol. 133, no. 9, pp. 690–700, https://doi.org/10.1061/(ASCE)0733-9364(2007)133:9(690).

Smith, P 2014, 'BIM & the 5D project cost manager', *Procedia – Social and Behavioral Sciences*, vol. 119, pp. 475–484, https://www.sciencedirect.com/science/article/pii/S1877042814021442.

Standards Australia 2011, *Cranes, hoists and winches – safe use – general requirements*, AS 2550.1-2011, Standards Australia, Sydney.

State Environmental Planning Policy (Transport and Infrastructure) 2021 (NSW), https://legislation.nsw.gov.au/view/html/inforce/current/epi-2021-0732.

Transport for NSW 2022, *Digital engineering framework*, version 4.0, TfNSW, Sydney, https://www.transport.nsw.gov.au/system/files/media/documents/2022/Digital-Engineering-Framework-v4.0.pdf.

Work Health and Safety Act 2011 (NSW), https://legislation.nsw.gov.au/view/html/inforce/current/act-2011-010.

Work Health and Safety Regulation 2017 (NSW), https://legislation.nsw.gov.au/view/html/inforce/current/sl-2017-0404.